KOÇ UNIVERSITY
College of Engineering

Interactive Ultrasonic Theremin

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Project Description

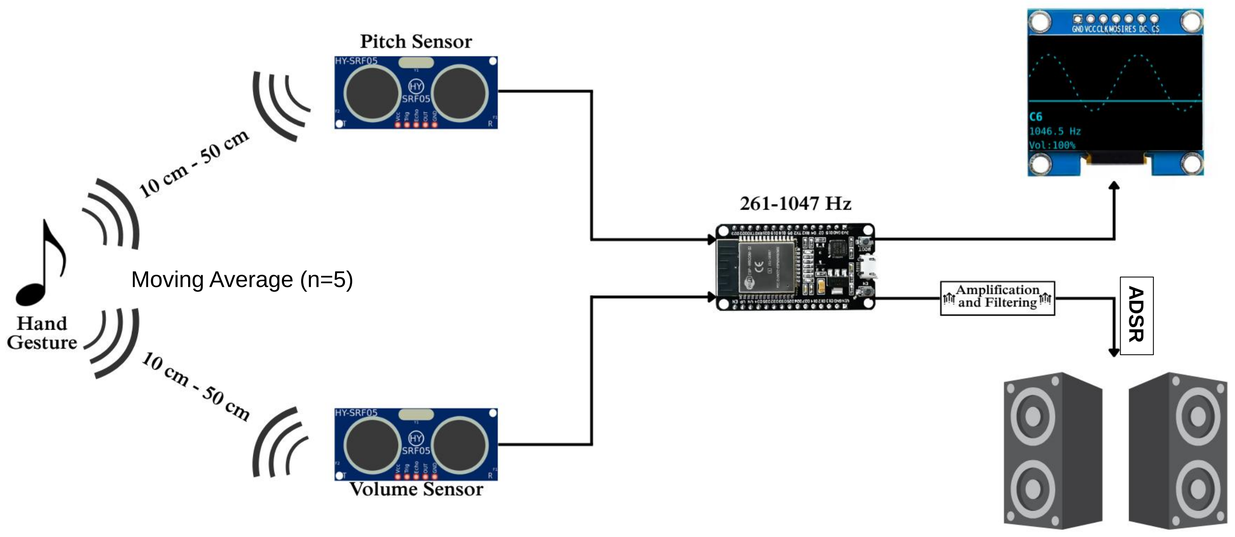


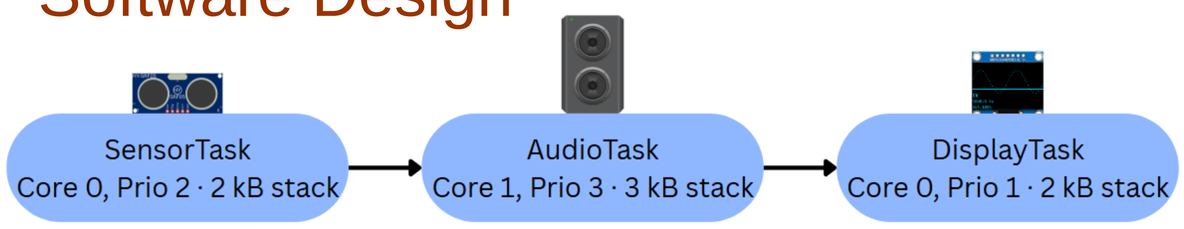
Figure 1: Summary of the System

- **Goal:** To build a stand-alone, contact-free musical instrument that converts hand distance into pitch & loudness in real time.

Sensing (5 kHz)	Trigger 10 µs
Mapping (1 kHz)	Map distance→note index (C4–C6)
Synthesis (44 kHz)	DDS sine lookup (256 samples)
UI (20 fps)	Waveform redraw
Total latency ≈ 43 ms	

Figure 2: Signal Flow / Timing Diagram

Software Design



- Three cooperative tasks share data via queues, guaranteeing < 50 ms end-to-end latency even under Wi-Fi debug logging.

Moving Average

$$\bar{d}[\mathbf{n}] = \frac{1}{5}\sum_{k=0}^{4} d[n-k]$$

ADSR

Attack 5 ms · Decay 50 ms
Sustain var · Release 80 ms

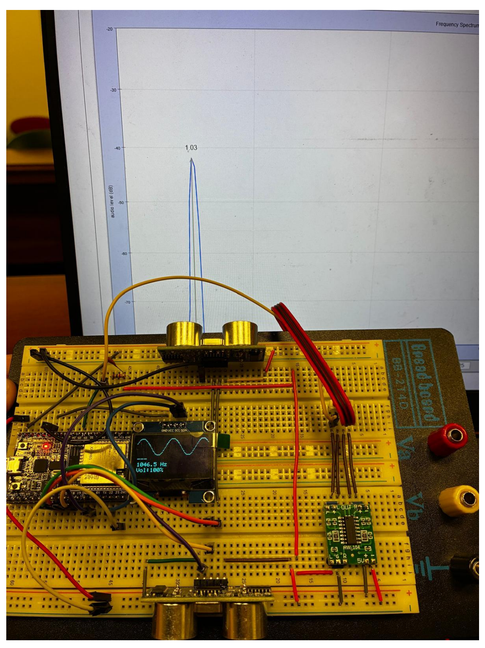
Figure 3: Smoothing Algorithms

- The firmware is built on FreeRTOS 10.5 (ESP-IDF v 5.3).
  SensorTask triggers both HY-SRF05 modules, applies a 5-tap MA followed by an EMA to reject mis-echoes, and maps distance to pitch & amplitude.
  AudioTask runs a 256-entry DDS oscillator at 44 kHz; the 8-bit result is fed to the ESP32's new **SDM** (noise-shaped 1-bit) peripheral, yielding 8 dB lower hiss than PWM.
  DisplayTask redraws a waveform + note text at 20 fps using lcdgfx.
  The three tasks communicate through lock-free queues, giving 43 ms worst-case latency and 37 % peak CPU use.

Group Members' Photo



Applications



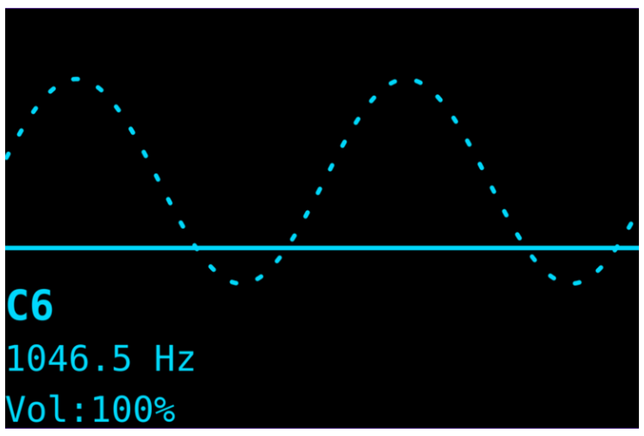
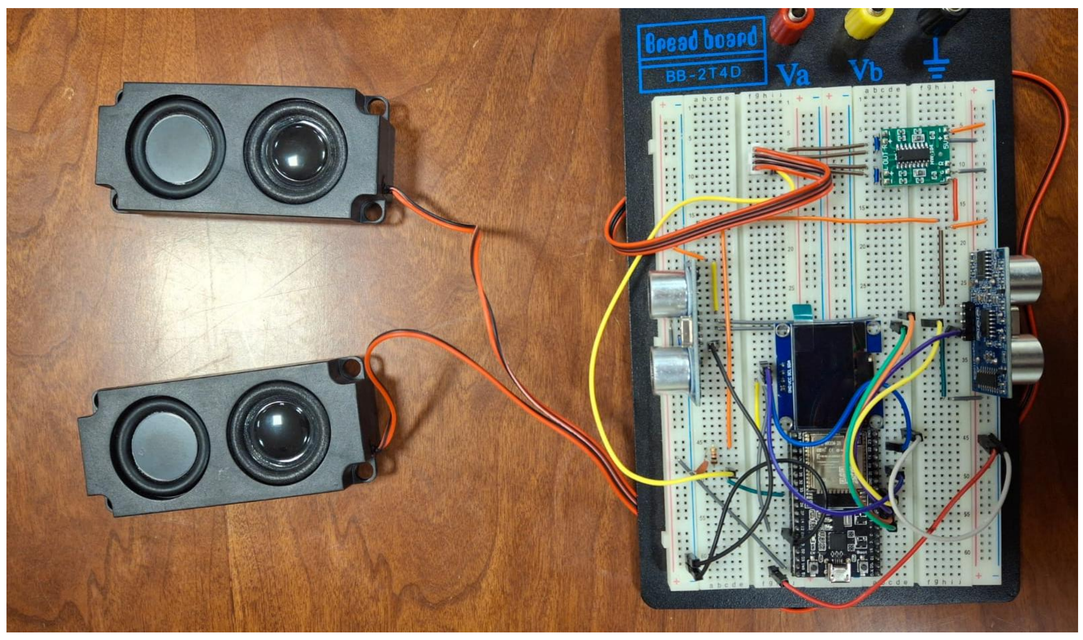


Figure 4: Applications: a. Output of Desired Note b. Oled Display Screen

Hardware Design



- A single-rail **5 V** bus powers the ESP-WROOM-32U module, dual HY-SRF05 sensors, SH1106 OLED and the PAM8403 class-D amplifier. Audio is cleaned in **two stages**: a passive **RC low-pass (3 kΩ + 6.8 nF, fc ≈ 7 kHz)** straight after the ESP32's sigma-delta pin, followed—optionally—by a 2-pole active filter on the PCB for studio-quiet output. Each high-current node carries local bulk decoupling (**100 µF || 0.1 µF** at the PAM, 47 µF at ESP VIN) to eliminate supply dip-induced drop-outs.

[1] HY-SRF05 Ultrasonic Ranger Datasheet, micros.com.pl [2] Espressif ESP-WROOM-32U Datasheet, Espressif, 2022
[3] lcdgfx Library, https://github.com/lexus2k/lcdgfx [4] PAM8403 3 W Class-D Audio Amplifier Datasheet, Diodes Inc., 2021
[5] ADSR Envelope, https://github.com/mohayonao/adsr-envelope